

Evergreen Market Data Analytics Project Proposal

Introduction

Background: Evergreen Market is a nationwide retail chain with over 350 physical stores with a rapidly growing online presence. The company offers a wide range of products, including pantry staples, personal care items, seasonal merchandise, and home essentials. (Evergreen Market scenario, Southern New Hampshire University, 2024) This retail chain is facing inventory challenges. These challenges are creating an increase in costs, reducing customer satisfaction, and creating inefficiencies. This company’s current analytics approach is mostly reactive. To better align inventory with customer demand Evergreen needs more advanced predictive tools.

Data Sources: Evergreen Market’s dataset contains 73,100 entries across 15 variables. The market data includes essential variables, but because of what I generated below such as boxplots, histograms, and summary statistics it shows significant variability, skewed demand forecasts, and multiple outliers. To me this indicates that the data alone is not reliable enough for accurate predictive decision-making. This data is useful for understanding past performance. Without higher quality data inputs, the dataset is not adequate for precise forecasting and optimizing inventory.

In [1]: import pandas as pd; df = pd.read_csv('retail_store_inventory.csv'); df.head()

Out[1]:

Category	Region	Inventory Level	Units Sold	Units Ordered	Demand Forecast	Price	Discount	Weather Condition	Holiday/Promotion	Competitor Pricing	Seasonality
Groceries	North	231	127	55	135.47	33.50	20	Rainy	0	29.69	Autumn
Toys	South	204	150	66	144.04	63.01	20	Sunny	0	66.16	Autumn
Toys	West	102	65	51	74.02	27.99	10	Sunny	1	31.32	Summer
Toys	North	469	61	164	62.18	32.72	10	Cloudy	1	34.74	Autumn
Electronics	East	166	14	135	9.26	73.64	0	Sunny	0	68.95	Summer

In []:

Ryan Bruno [25-Jul-2026 09:10:17 PM]
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[0Cpmg-09JxQ-08KPY]

In [1]: import pandas as pd; df = pd.read_csv('retail_store_inventory.csv'); df.head()

Out[1]:

	Date	Store ID	Product ID	Category	Region	Inventory Level	Units Sold	Units Ordered	Demand Forecast	Price	Discount	Weather Condition	Holiday/Promotion
0	2022-01-01	S001	P0001	Groceries	North	231	127	55	135.47	33.50	20	Rainy	
1	2022-01-01	S001	P0002	Toys	South	204	150	66	144.04	63.01	20	Sunny	
2	2022-01-01	S001	P0003	Toys	West	102	65	51	74.02	27.99	10	Sunny	
3	2022-01-01	S001	P0004	Toys	North	469	61	164	62.18	32.72	10	Cloudy	
4	2022-01-01	S001	P0005	Electronics	East	166	14	135	9.26	73.64	0	Sunny	

In []:

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In [2]: df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 73100 entries, 0 to 73099
Data columns (total 15 columns):
#   Column              Non-Null Count  Dtype  
---  --
0    Date                73100 non-null  object  
1    Store ID            73100 non-null  object  
2    Product ID          73100 non-null  object  
3    Category            73100 non-null  object  
4    Region              73100 non-null  object  
5    Inventory Level     73100 non-null  int64   
6    Units Sold          73100 non-null  int64   
7    Units Ordered       73100 non-null  int64   
8    Demand Forecast     73100 non-null  float64  
9    Price               73100 non-null  float64  
10   Discount            73100 non-null  int64   
11   Weather Condition   73100 non-null  object  
12   Holiday/Promotion    73100 non-null  int64   
13   Competitor Pricing  73100 non-null  float64  
14   Seasonality          73100 non-null  object  
dtypes: float64(3), int64(5), object(7)
memory usage: 8.4+ MB
```

Ryan Bruno [25-Jul-2026 09:12:42 PM]
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[OCpmg-09JxQ-08KrY]

In [3]: df.describe()

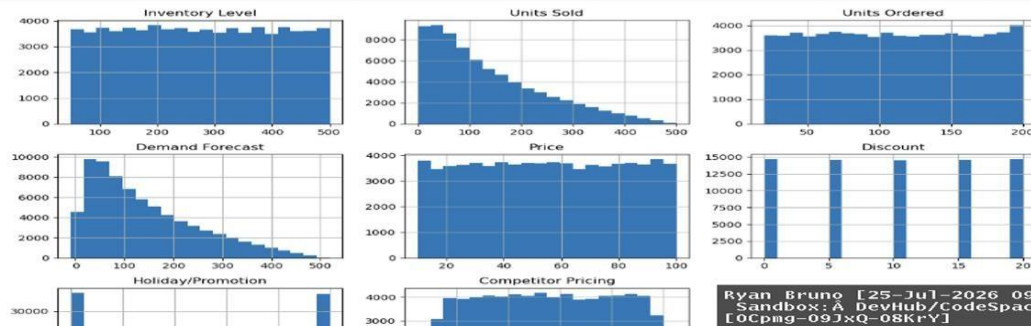
Out[3]:

	Inventory Level	Units Sold	Units Ordered	Demand Forecast	Price	Discount	Holiday/Promotion	Competitor Pricing
count	73100.000000	73100.000000	73100.000000	73100.000000	73100.000000	73100.000000	73100.000000	73100.000000
mean	274.469877	136.464870	110.004473	141.494720	55.135108	10.009508	0.497305	55.146077
std	129.949514	108.919406	52.277448	109.254076	26.021945	7.083746	0.499996	26.191408
min	50.000000	0.000000	20.000000	-9.990000	10.000000	0.000000	0.000000	5.030000
25%	162.000000	49.000000	65.000000	53.670000	32.650000	5.000000	0.000000	32.680000
50%	273.000000	107.000000	110.000000	113.015000	55.050000	10.000000	0.000000	55.010000
75%	387.000000	203.000000	155.000000	208.052500	77.860000	15.000000	1.000000	77.820000
max	500.000000	499.000000	200.000000	518.550000	100.000000	20.000000	1.000000	104.940000

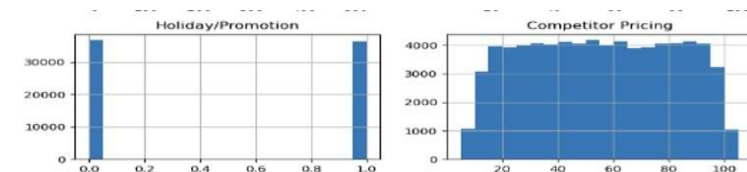
In []:

Ryan Bruno [25-Jul-2026 09:13:32 PM]
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[OCpmg-09JxQ-08KrY]

In [4]: import matplotlib.pyplot as plt; df.hist(figsize=(12,8), bins=20); plt.tight_layout(); plt.show()

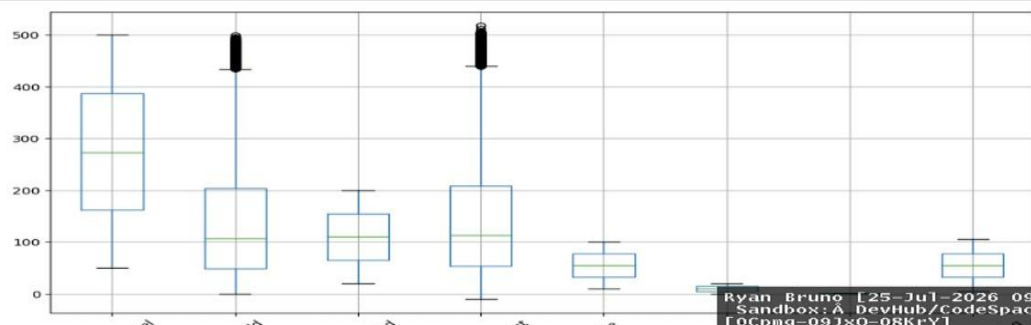


Ryan Bruno [25-Jul-2026 09:17:41 PM]
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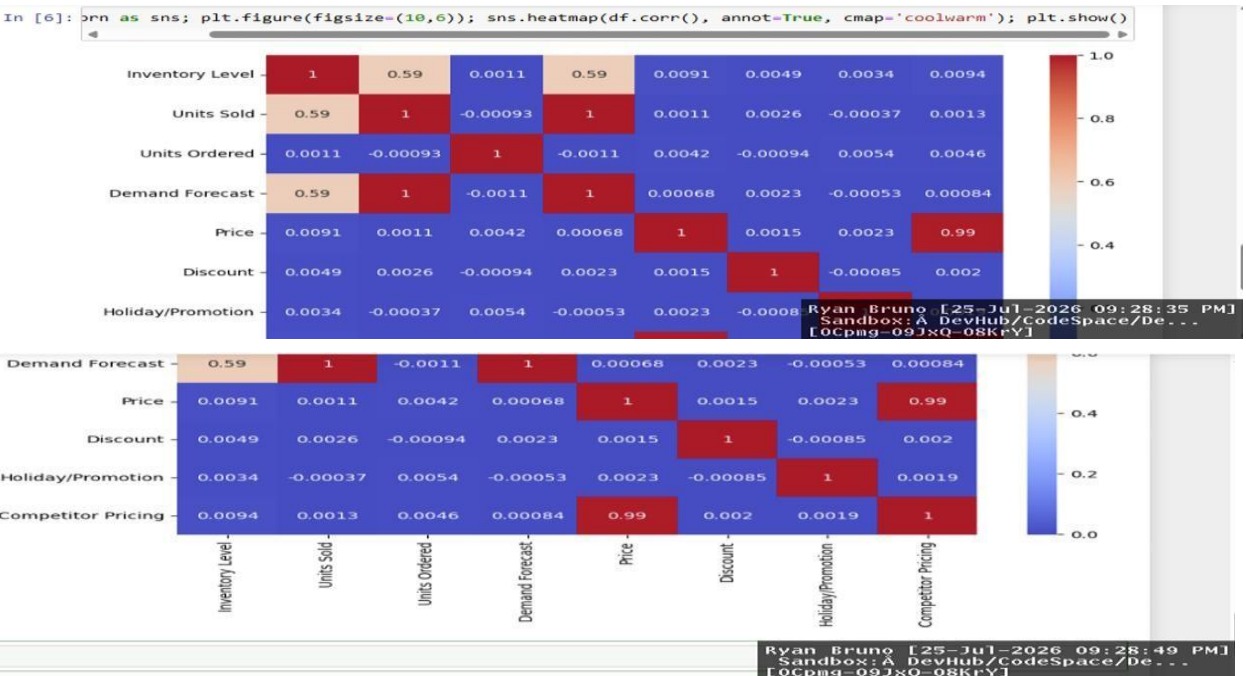
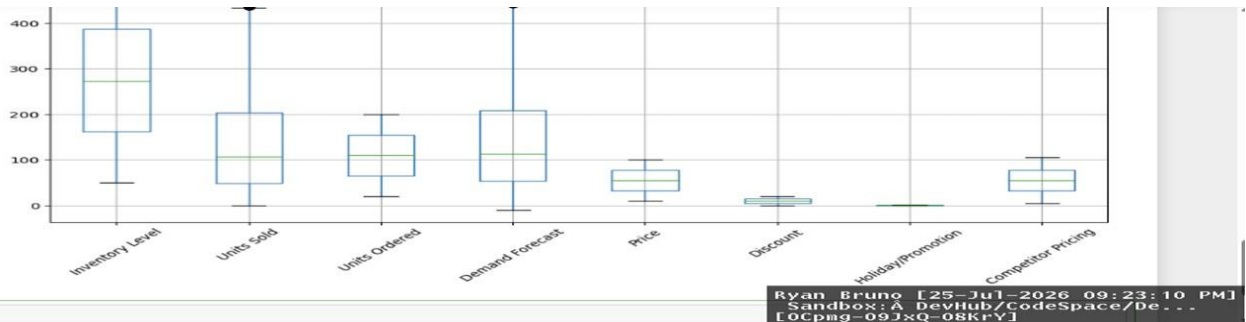
Ryan Bruno [25-Jul-2026 09:17:55 PM]
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In [5]: plt.figure(figsize=(12,6)); df.boxplot(); plt.xticks(rotation=45); plt.show()



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Data Needs: The current data set captures metrics such as inventory levels, sales, pricing, and demand forecasts but it lacks external/behavioral data which is needed for more accurate predictions. Focusing on customer purchasing patterns, regional events, and online browsing behavior would explain demand fluctuations that existing data cannot fully capture. Additional data sources like these should give Evergreen a more complete view of what is driving demand. Once Evergreen notices what drives demand it will support inventory optimization.

Data Analytics Initiative: To support proactive inventory optimization Evergreen should be adding customer purchasing patterns, regional events, and online browsing behavior to their data analysis. This would help explain demand fluctuations that current data can't fully capture.

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Proposal

Goals: Evergreen Market's goals are to identify top performing and underperforming products categories, and ensure data quality by detecting missing values, duplicates, and outliers. Sales trends need to be tracked over time to support inventory planning. As mentioned earlier, there is mounting operational pressure and that is why Evergreen Market initiated a formal data analytics program. Evergreen has a broader mission of reducing operational waste, improving profitability, and delivering consistent performance across all regions. If all these goals are complete Evergreen's broader mission will be accomplished; Efficiency will be maximized in every sense of the word. Evergreen Market will be able to reduce operational waste after analyzing both under and over performing products on a graph/chart format which I will provide below. Overall success will be measured by tracking key performance indicators such as profit margin improvement, a reduction in loss generated discount transactions, and improved data completeness score. Data evaluation and Exploratory Data Analysis will be conducted as we progress through the project proposal.

Data Analytics Life Cycle: The Data analytics life cycle is a step-by-step process which helps from initial identification of a business problem all the way through to data collection, cleaning, exploration, modeling, and then communicating all findings to stakeholders. By following this Life cycle, Evergreen will ensure every stage of the analysis is connected to our business goals.

Phase 1) Business Problem: This phase helps establish the current problem(s) the business has and the appropriate analysis method to use. A problem with Evergreen's market is profit margins are declining across certain regions as well as product categories.

Phase 2) Data Collection: Data collection gives a complete view of how Evergreen is performing across products, locations, time, etc. The data set used for this initiative contains

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Evergreen Market Project Proposal (Milestone Two) Evergreen Market's transactional sales records include fields such as Product Category, Sales, Profit, Discount, Region, and Order Date.

Phase 3) *Data processing/transformation and data cleaning:* The data must be verified for quality before any analysis; three checks are performed to ensure the dataset is reliable.

Missing value check: Columns with zero missing values indicate reliable data collection

```
In [6]: missing_values = df.isnull().sum()
print(missing_values)

Date                0
Store ID            0
Product ID          0
Category            0
Region              0
Inventory Level      0
Units Sold          0
Units Ordered        0
Demand Forecast      0
Price               0
Discount             0
Weather Condition    0
Holiday/Promotion    0
Competitor Pricing   0
Seasonality          0
dtype: int64
```

Duplicate Records Check: Confirming zero duplicates supports data reliability

```
In [7]: duplicate_rows = df.duplicated().sum()
print("Number of duplicate rows:", duplicate_rows)

Number of duplicate rows: 0
```

Outlier Detection Using IQR: Excessive outliers indicate inconsistent data entry

```
In [8]: numeric_cols = ['Inventory Level', 'Units Sold', 'Units Ordered', 'Demand Forecast', 'Price']

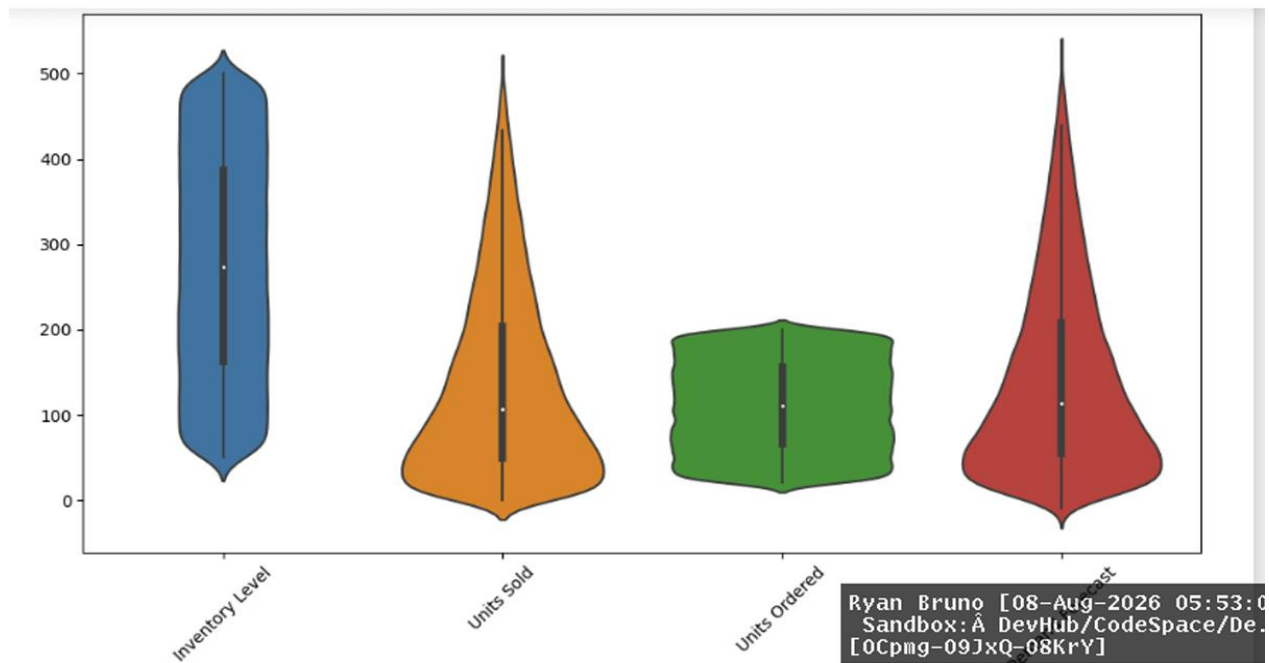
for col in numeric_cols:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1
    outliers = df[(df[col] < Q1 - 1.5*IQR) | (df[col] > Q3 + 1.5*IQR)]
    print(f"{col}: {len(outliers)} outliers")

Inventory Level: 0 outliers
Units Sold: 715 outliers
Units Ordered: 0 outliers
Demand Forecast: 732 outliers
Price: 0 outliers
```

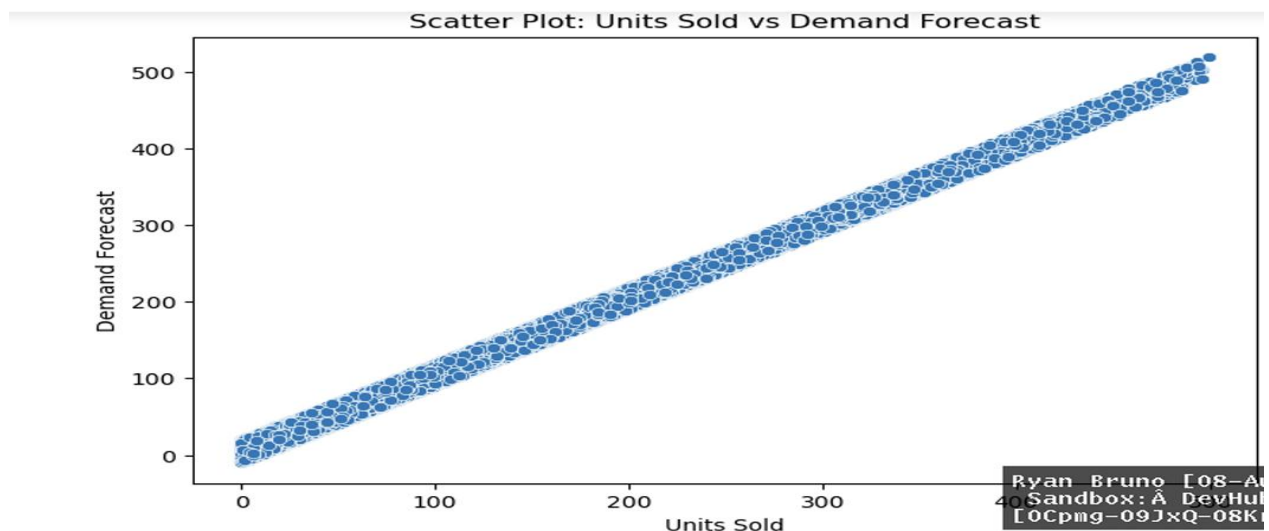
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Phase 4) Exploratory Data Analysis (EDA): Now that clean data is in hand the analysis uses multiple visualizations to surface patterns. It can help define earlier business problems.

Violin Plots of key analytical variables (5.8.1) (Demand Forecast is red):

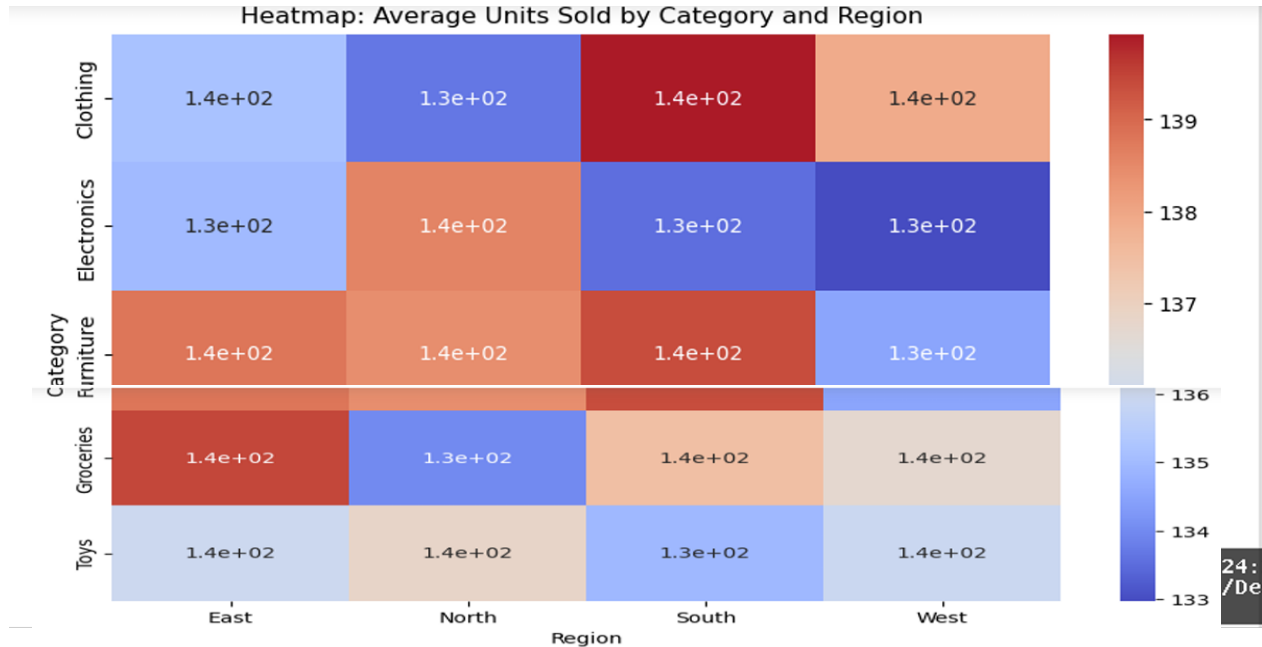


Scatter Plot: Units Sold Vs Demand Forecast (5.8.2):



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Heatmap: Average Units Sold by Category and Region (5.8.4):



Phase 5) Modeling and validation: Patterns uncovered during EDA will be formalized using predictive techniques such as regression analysis. This is what will quantify the exact impact of discounting on profits.

Phase 6) Visualization with proposed action: This phase determines the best insights based on analysis. After that, decisions on how we should hit our goal will be made. From the pivot heatmap, it shows that sales volume and profit margins are already strong. This tells me additional support is most likely to generate a measurable return. I propose that Leadership should redirect promotional investment toward the highest-performing category-region combinations.

Value of Life Cycle: This Data Analytics Life Cycle strengthens Evergreen's Market initiative by improving predictability, performance, and security through a structured approach. Each phase

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makes sure the data is clean and relevant which leads to reliable insights. Evergreen can trust its results are based on validated data.

Data: Evergreen's market data set is useful because key operational variables are included. This is needed for trend analysis; it is complete and free of duplicates. Outliers on the other hand limit predictive accuracy which means additional data sources would strengthen the initiative. New Data collected must be securely managed to maintain compliance with privacy standards.

Tool Applicability to Initiative: The current analytics tools of Evergreen Market are primarily descriptive and reactive. This limits their ability to support your proactive inventory optimization initiative. My Milestone One analysis shows that Evergreen relies on basic reporting and summary statistics to understand inventory levels, sales, pricing, and demand forecasts (Evergreen Market Dataset). These tools cannot support advanced forecasting or behavioral analysis, but they are effective for monitoring past performance. The initiative requires identifying Demand drivers and predicting future inventory needs. The existing tools cannot perform clustering not can they integrate new data sources such as online browsing behavior. As noted, "the data alone is not reliable enough for accurate predictive decision-making" this is because of variability, skewed forecasts, and outliers (Milestone One). Therefore, while Evergreen's current tools provide a foundation for descriptive analytics, they are not sufficient for the predictive and integrative capabilities required for this initiative.

Tool Applicability to Data: The existing tools work well with Evergreen's structured transactional dataset. Variables are included such as Inventory Level, Units Sold, Units Ordered, Price, Discount, and Seasonality. Histograms, boxplots, and summary statistics are all shown in Milestone One demonstrating their compatibility with structured data. Although this is the case, the dataset also contains significant variability and hundreds of outliers. Most of these outliers are in Units Sold and Demand Forecast to be more specific. Basic tools can detect the outliers, but

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they will not be modeled effectively. My work in Milestone two further confirms this, noting “Units Sold: 715 outliers” and “Demand Forecast: 732 outliers,” which limits predictive accuracy significantly. Additionally, the initiative requires new forms of data. Some forms of data we can use are customer purchasing patterns, regional events, and online browsing behavior. These might be unstructured and Evergreen’s current tools cannot process or analyze them. To summarize, Evergreen’s existing tools are appropriate for structured data it currently has but they are not sufficient for the expanded dataset required for accurate forecasting.

Tool Recommendations: I have two tools for recommendation that could be reasonably applied to this initiative- Tableau or Power BI and Python with Scikit-Learn. Tableau/ Power BI is used for advanced visualization and dashboarding; Python with Scikit-Learn is a machine learning toolkit.

- 1) Tableau or Power BI: Currently Evergreen is using basic reporting tools, but the initiative demands ongoing monitoring of trends and regional performance. Tableau/ Power BI can connect to Evergreen’s structured data. It can also add any new data sources later. The tools provided by Tableau/ Power BI support interactive dashboards, real-time KPI tracking, and clear communications of insights. These are all essential components of Phase 6 of the Data Analytics Life Cycle (Milestone Two). This helps leadership track profit margin improvements, and it visualizes underperforming as well as overperforming product categories. By adopting Tableau or Power BI, Evergreen’s broader mission of removing waste and improving profitability becomes more attainable.
- 2) Python with Scikit-Learn: Python I have already been using in my Jupyter Notebook work, but Evergreen Market does not currently use it as an enterprise analytics tool. Scikit-Learn would allow Evergreen to build predictive models - regression, clustering, demand forecast – all essential for the initiative. This tool can handle large datasets and

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since there is a 73,100-row dataset we are dealing with it would fit perfectly. This tool also enables anomaly detection to flag unusual inventory behavior. This would improve accuracy and reduce operational waste.

Summary: The tools that Evergreen are currently using are adequate for descriptive analysis but insufficient for predictive modeling. These existing tools work well with structured transactional data, but it can't handle outliers, variability, or massive Data Sets. Evergreen will have the ability to forecast demand accurately and support proactive inventory optimization if we start utilizing Python with Scikit-Learn and Tableau/Power BI.

Conclusion

Value: The application of data analytics to Evergreen Market delivers substantial value by transforming Evergreen's inventory management from a reactive process into a data-driven strategy. The analyses showed Evergreen's current approach which relies heavily on descriptive reporting and basic summary statistics. This cannot support accurate forecasting or explain demand fluctuations. The initiative proposed which incorporates additional behavioral and external data provides Evergreen with a more complete, actionable understanding of what drives demand. The existing dataset of 73,100 records reveals significant variability/ hundreds of outliers in key fields - Units sold; Demand Forecast – limiting predictive accuracy. Adopting advanced analytics tools such as Python with Scikit-Learn, Tableau, or Power BI can improve. Evergreens forecast precision, reduce operational waste, and optimize inventory levels. With these tools leadership can now monitor KPIs in real time while also allowing leadership to identify underperforming and overperforming product-region combinations. Leadership can now make strategic decisions which support Evergreen's broader mission of improving profitability and reducing inefficiencies. Across all regions, data analytics enable Evergreen Market to meet

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its goals by improving demand forecasting, strengthening inventory alignment, reducing unnecessary discounting, and enhancing operational consistency. Actionable insights derived from data analytics support Evergreen's long-term growth and stability within the retail industry.

Insights: Insights gained from my analysis of Evergreen Market highlight the transformative potential of data analytics. The Exploratory Data Analysis (EDA) revealed clear patterns. These patterns that have become clearer due to EDA are strong category-region performance combinations and inconsistencies between demand forecasts and actual sales. Leadership can use these clear patterns to improve inventory planning. The heatmap visualizations and scatter plots demonstrated how data analytics can uncover relationships that basic reporting tools cannot. This reinforces the need for more sophisticated modeling techniques. Evergreen's current tools are insufficient for handling large datasets. Predictive models – regression and clustering - would allow Evergreen not only to quantify demand drivers but to forecast future inventory needs with greater accuracy. These insights are beneficial to Evergreen, but they also reflect broader industry trends; retailers that adopt predictive analytics are better equipped to manage supply chain disruptions and meet evolving customer expectations. For me, personally, this project strengthened my understanding of the Data Analytics Life Cycle while also enhancing my technical skills in Python, data cleaning, visualization, model evaluation, and so much more. I am beyond grateful that I worked on this project, specifically looking into Evergreens Market data. Insights I have gained from this project will support my future growth; as a future Analytics professional my analytical skills across the board have improved. I had exposure to many different applications and graphs that I have not used or seen before; Violin Plots, Pivot-Based Heatmaps, Correlation Heatmaps are all new to me. Knowing how to code to create these plots for a specific market will help tremendously with my future career as a Data Analyst.

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Communication

Visualization: Each chart was selected to highlight patterns which directly impact profitability, inventory alignment, and operational efficiency. The **histograms and box plots** reveal distribution and variability of key variables – Inventory Level, Units Sold, Units Ordered, Demand Forecast, and Price. These visuals make it clear that Evergreen’s dataset contains multiple extreme values (ex. “Units Sold: 715 outliers” and “Demand Forecast: 732 outliers”). This tells us we need improved forecasting tools. The **Violin Plot** deepens this understanding; it shows density and variability across categories. Now leadership can quickly see where demand is inconsistent. A major operational issue is highlighted with a visual of the **scatter plot of Units Sold vs. Demand Forecast**. This shows that forecasts often fail to match actual sales. Even without technical interpretation this misalignment is visible. A high-level view of relationships between variables is shown from the **correlation heatmap**. While correlations are generally weak, the heatmap visually communicates Evergreen’s current forecasting inputs lack strong predictive power. This justifies the need for expanding the dataset with behavioral and external variables. I created another heatmap not to be confused with the correlation heatmap – the **pivot-based heatmap of Average Units Sold by Category and Region**. This heatmap offers a clear picture of where performance is strongest. This visualization directly supports strategic decision-making with high-performing category-region combinations that deserve increased promotional investment are being identified. All visualizations communicate Evergreen’s challenges and opportunities in a way that is aligned with management’s decision-making needs.

Presentation: The report begins by outlining Evergreen’s core challenge: inventory misalignment driven by reactive analytics and unreliable demand forecasts. Visual evidence is then used to demonstrate how variability, outliers, and weak correlations limit the accuracy of

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current forecasting methods. As noted earlier, “The data alone is not reliable enough for accurate predictive decision-making.”

The proposal emphasizes business outcomes rather than technical mechanics. For example:

1. Better inventory alignment lowers operational waste
2. Identifying strong category(s) (region combinations supports targeted promotions)
3. Integrating new data sources strengthens predictive accuracy
4. Improved demand forecasting reduces unnecessary discounting

I am guiding Leadership toward a clear conclusion: adopting advanced analytics tools (Python with Scikit-Learn and Tableau/ Power BI) will transform Evergreen’s inventory management from reactive to proactive.

Final Recommendations for Evergreen Market

- Adopt advanced predictive modeling tools
- Implement Tableau or Power BI dashboards
- Expand data collection (behavioral and external variables)
- Prioritize data cleaning as well as preprocessing workflows
- Redirect promotional investments
- Develop anomaly detection systems
- Establish a continuous analytics improvement cycle

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